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Spring 2025 ARML Power Contest

The Background

As mathletes, many of you are familiar with the Fibonacci sequence: 1, 1, 2, 3, 5, 8, 13, 21, The sequence starts with $F_1 = 1$ and $F_2 = 1$ and then each term after that is the sum of the two previous terms, so that $F_5 = F_3 + F_4 = 2 + 3 = 5$, for instance. The sequence can be run backwards as well, by reversing the relationship to read $F_3 = F_5 - F_4$ so that $2 = 5 - 3$. In this way, $F_0 = F_2 - F_1 = 1 - 1 = 0$, $F_{-1} = F_1 - F_0 = 1 - 0 = 1$ and so on.

For this contest, you will work with *Fibonacci-like sequences*, sequence that satisfy the following conditions:

1. Each term is the sum of the previous two terms.
2. The terms (of positive index) are positive and nondecreasing.

The first condition is called the *Fibonacci recursion*. The sequence $F(a, b)$ will be the sequence starting $a, b, a + b, a + 2b, 2a + 3b$, and continuing with the add-two-to-get-the-next-term pattern. For instance, $F(2, 5)$ is the sequence 2, 5, 7, 12, 19, 31, 50, You may refer to individual terms in such a sequence by using their index as a subscript: $F(2, 5)_6 = 31$ for example. Note that the nondecreasing condition requires $a \leq b$. If you need to go backward to zero or negative indices, the second condition need not hold. For example, $F(2, 5)_0 = 5 - 2 = 3$, even though $3 > 2$ so that the series fails to be nondecreasing.

Note that the regular Fibonacci sequence is $F(1, 1)$. It may be referred to in shorter form, without reference to $a = 1$ and $b = 1$. That is, you may write $F_6 = 8$ instead of having to write $F(1, 1)_6 = 8$.

Your main challenge will be to find a “small” Fibonacci-like sequence that contains a given positive integer. It should be clear that any positive integer n occurs in *some* Fibonacci-like sequence. In fact, $F(n, b)$ starts $n, b, n + b, \dots$ and contains n (as its first term!) for any choice of $b \geq n$. Or, you could make the first term of a sequence containing n be as small as possible: $F(1, n - 1)$ is the sequence 1, $n - 1, n, \dots$. Since both of these are rather trivial, what should “small” mean? The “smallest” Fibonacci-like sequence containing a given number n will be $F(a, b)$ with the smallest possible *second* number b that contains n . Define the function $B(n)$ to be the smallest b for which there is a Fibonacci-like sequence $F(a, b)$ that contains n . (You will show that each n has a unique smallest b and only one choice of a works for that b .) For example, $F(1, 49)$ contains 50 as its third term. $F(4, 23)$ contains 50 as its fourth term. But $F(2, 5)$ contains 50 as its seventh term, and no smaller choice of b produces a sequence that contains 50, so $B(50) = 5$.

The Problems

There are 40 points available on this contest. The numbers at the end of each problem are the number of points that problem is worth. Be sure to read the problems carefully and answer the question that is being asked!

1. For $F(3, 10)$, compute the first 10 terms. Also, compute the terms $F(3, 10)_0$ and $F(3, 10)_{-1}$. [3 pts.]
2. Show, by writing out all possible cases, that $B(50) = 5$ and that there is only one Fibonacci-like sequence with $b = 5$ that contains the number 50. [3 pts.]
3. Showing work, compute $B(22)$. [2 pts.]
4. Note that the general Fibonacci-like sequence $F(a, b)$ begins $a, b, a + b, a + 2b, 2a + 3b, 3a + 5b, 5a + 8b, \dots$. You might recognize those coefficients of a and b as the regular Fibonacci numbers. Prove that for $k \geq 3$ that $F(a, b)_k = F_{k-2}a + F_{k-1}b$. (This relationship actually holds for all integers k , by running the sequences backward as explained in the background material.) [3 pts.]
5. Prove that if $F(a_1, b)_k = F(a_2, b)_k$ then $a_1 = a_2$. That is, if the same number occurs in the same location in two Fibonacci-like sequences with the same b value, they must be the exact same sequence. [2 pts.]
6. Prove that for any integers $1 \leq j < k$ that $\frac{F_k - F_{k-j}}{F_{k-j-1}} \geq 1$, with equality if and only if $j = 1$. [2 pts.]
7. Let $F(a, b)_k = F(a', b')_{k-j}$ for some $1 \leq j < k$. That is, the same number appears in both Fibonacci-like sequences $F(a, b)$ and $F(a', b')$ but appears earlier in the latter. Prove that $b < b'$. [4 pts.]

Problem 7 shows that for Fibonacci-like sequences containing n , the one with the smallest value of b is also the *longest*, that is, it contains the most terms before reaching a value of n . Problem 5 then shows that if two sequences with the same b contain the same number in the same place, they are the same sequence. This completes the uniqueness proof referred to in the background material. In fact, these problems show that the sequences $F(a, b)$ and $F(a', b)$ with $a \neq a'$ have only the number b in common, for if the number n occurred in the same place in both sequences, problem 5 would guarantee that $a = a'$ while if n occurred in different places in the two sequences, the longer one would have a smaller b .

Thus, in order to find $B(n)$ it seems necessary to find a way to modify Fibonacci-like sequences in order to make them longer, if possible.

Consider an example sequence, $F(19, 40)$:

19 40 59 99 158 267 425 692

Another way to have a sequence that contains the number 692 is if the terms leading up to it were $\dots 268, 424, 692, \dots$. The sequence can be *altered* by decreasing the penultimate term by 1 while simultaneously increasing the previous term by 1. What happens to the number previous to that one? The Fibonacci recursion allows it to be computed: $424 - 268 = 156$, so it appears that 2 is subtracted from that term. Then what?

Original:	19	40	59	99	158	267	425	692
Modified:	6	48	54	102	156	268	424	692
Differences:	-13	+8	-5	+3	-2	+1	-1	

The changes appear to be the regular Fibonacci sequence, with alternating signs.

8. Given a Fibonacci-like sequence $a_1, a_2, a_3, \dots, a_{k-1}, a_k$ that contains $a_k = n$. Let $b_1, b_2, \dots, b_k$ be another sequence of the same length, with $b_k = a_k = n$ that also satisfies the Fibonacci recursion, and that $b_{k-1} = a_{k-1} - 1$. Show that $b_{k-j} = a_{k-j} + (-1)^j F_j$. [2 pts.]

Note that the b -sequence is only required to satisfy the Fibonacci recursion; it may very well not satisfy the conditions on positivity or non-decreasing of its terms. For example, altering the above sequence by again subtracting 1 from the penultimate term would result in

$$-7 \quad 56 \quad 49 \quad 105 \quad 154 \quad 269 \quad 423 \quad 692$$

which has a negative term, and decreases from 56 to 49.

Alternatively, the penultimate term could be increased by 1 instead of decreased, and the changes propagate backward through the sequence, with the terms where j is odd *increasing* and the terms where j is even decrease.

Given a Fibonacci-like sequence, under some conditions the sequence can have more terms prepended to the beginning and still remain Fibonacci-like. For example, if the sequence begins $5, 9, \dots$, it could be *lengthened* by prepending 1, 4, and the new, longer sequence $1, 4, 5, 9, \dots$ is still Fibonacci-like.

9. Lengthen each Fibonacci-like sequence backward as far as possible while staying Fibonacci-like:

(a) $11, 17, \dots$ [1 pt.]

(b) $6, 8, \dots$ [1 pt.]

(c) $7, 16, \dots$ [1 pt.]

10. Given $a \leq b$, under what conditions on a and b may the sequence $a, b, \dots$ be extended backward

(a) by exactly one term? [2 pts.]

(b) by two or more terms? [2 pts.]

The goal is to find $B(n)$, which is the smallest second term of a Fibonacci-like sequence containing n . This, in turn, has been shown to be the sequence that takes the longest to reach n . The techniques of altering and lengthening sequences can be used in tandem. Start with the Fibonacci-like sequence $F(1, n-1) = 1, n-1, n, \dots$. Alter the sequence by decreasing the second number *until* the second technique can be applied to lengthen the sequence. Lengthen as much as possible and return to the process of altering the terms, always decreasing the penultimate term. When a sequence is reached that can neither be altered (by decreasing the penultimate term) nor lengthened, stop. $B(n)$ has been found!

Here is the technique applied to $n = 200$:

Start:					1	199	200
Alter:					2	198	200
Alter:					⋮	⋮	⋮
Alter:					67	133	200
Lengthen:		1	66	67	133	200	
Alter:		4	64	68	132	200	
Alter:		⋮	⋮	⋮	⋮	⋮	
Alter:		25	50	75	125	200	
Lengthen:	(0)	25	25	50	75	125	200
Alter:		20	28	48	76	124	200
Lengthen:	8	20	28	48	76	124	200
Alter:	16	15	31	46	77	123	200

The second-to-last row is the Fibonacci-like sequence that takes the longest to reach $n = 200$, and its second term is $B(200) = 20$. Note that this sequence cannot be altered or lengthened while remaining Fibonacci-like. Decreasing the penultimate term results in the final line of the table, where the sequence is not non-decreasing. And it has been lengthened as much as possible because it was arrived at by lengthening!

Algorithm for computing $B(n)$: Begin with the sequence $1, n-1, n$. Then

- Lengthen the sequence as much as possible, then continue to the next step.
- Attempt to alter the sequence by decreasing the penultimate term by 1.
 - If this alteration is possible, return to the previous (lengthen) step.
 - If this alteration is not possible, **STOP**. The current sequence is the Fibonacci-like sequence that takes as long as possible to reach n . $B(n)$ is the second term of this sequence.

11. Use this algorithm to find $B(199)$ and $B(201)$. [3 pts.]
12. Prove that algorithm is correct. That is, given any positive integer n , prove the algorithm always stops after a finite number of steps and produces the correct value of $B(n)$. [4 pts.]
13. Prove that if each sequence is lengthened as much as possible, then for each positive integer b there are exactly $1 + \lfloor (b-1)/2 \rfloor$ Fibonacci-like sequences with b as their second term. [2 pts.]
14. For each b in the range $1 \leq b \leq 10$, compute $L(b)$, which is the least number n such that $B(n) = b$. (Note: if you find that one of these values is larger than 100 you have made an error.) [3 pts.]

Much more is known about the functions $B(n)$ and $L(b)$, including a very nice pattern in the values of $L(b)$. If you are curious about these functions and want to learn more, seek out the article *A Small Fib* by John P. Bonomo and Montana Ferita in the *College Mathematics Journal*, 54:4 (September 2023).